LLM as a Judge: Evaluating AI with AI

Introduction to AI-Powered Assessment

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LLM-as-a-Judge: Overview

- **LLM-as-a-Judge** is an evaluation method used to assess the quality of text generated by LLM-based systems, such as chatbots, Q&A tools, or AI agents.
- Instead of relying only on fixed metrics (BERTScore, BLEU, ROUGE), this approach uses another language model as a “judge” to evaluate outputs based on criteria that **you** define.
- You take the output of your model and pass it to another LLM together with an “evaluation” **prompt.**

LLM-as-a-Judge: Overview

- You can evaluate many different aspects of text, such as:
  - **Politeness:** Is the answer respectful?
  - **Bias:** Does it show unfair assumptions?
  - **Tone:** Is it formal or conversational?
  - **Sentiment:** Is it positive, negative, or neutral?
  - **Hallucinations:** Does it stay grounded in the provided context?

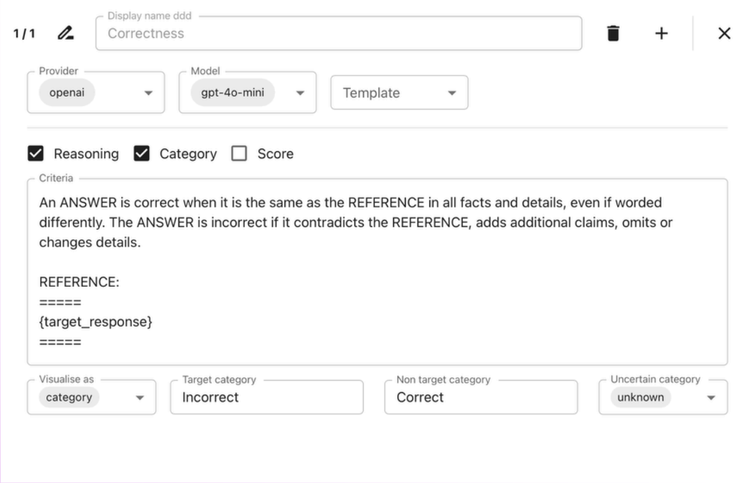
LLM-as-a-Judge: Overview

- The method also supports different evaluation styles:
  - For example, you can provide extra context, like a source document or reference, and ask the LLM to score the response **(reference-based evaluation):**
    - **Answer + Question:** Common in chatbots and Q&A systems to check if the response properly addresses the question etc.
    - **Answer + Reference Answer:** Helpful if you have a correct answer to compare against.
      - In my case, I used the “correct_answers” from my ground truth dataset as a reference for evaluation.

Why use LLM-as-a-Judge?

- It's a flexible technique for assessing outputs in a human way, without requiring costly human time.

How it works in practice



To apply this method, you usually:

1. **Define your evaluation criteria**
2. **Create a small dataset to test your LLM judge**
3. **Label the dataset manually, just as you want the LLM judge to do later**
   - This labeled dataset will be your "ground truth" and help measure how well the LLM judge performs → This may be different for your use case! You can, of course, skip this step and write the best prompt you can.
4. **Write an evaluation prompt**
5. **Set a low temperature**
   - In LLMs, temperature controls how random the output is. You don't need creativity for evaluation, therefore set the temperature low.
6. **Let the LLM judge evaluate the outputs**
7. **Use the results to improve your system**

Limitations and Challenges

- Even though this method is powerful, it also has some **challenges.**
  - The results are not always deterministic, and they depend heavily on:
    - the evaluation prompt,
    - the judge model you choose, and
    - the complexity of the task.
  - Because of this, building a good LLM judge is an iterative process.
    - You often need to refine the evaluation prompt multiple times to get reliable results.
      - In a way, the evaluation system itself needs to be evaluated and improved.